7. Glossary of AI and Machine Learning Terms

From the Chain Rule to ChatGPT: The Mathematics Behind AI

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How to use this glossary. Terms are grouped by topic and ordered to build on each other within each section. If a term in a definition is itself defined elsewhere in the glossary, it appears in **bold** on first use within that definition.

Hardware and Infrastructure

GPU (Graphics Processing Unit). A processor originally designed for rendering graphics, now the standard hardware for training neural networks. GPUs excel at performing thousands of simple arithmetic operations simultaneously — a property called *parallelism* — which makes them ideal for the matrix multiplications that dominate neural network computation. A single high-end GPU can perform hundreds of trillions of floating-point operations per second.

TPU (Tensor Processing Unit). A processor designed by Google specifically for neural network computation. Similar in purpose to a GPU but optimized for the particular arithmetic patterns of deep learning. GPT-3 was trained on thousands of GPUs; Google's models are typically trained on thousands of TPUs.

Training vs. Inference. *Training* is the process of adjusting a model's parameters to minimize a loss function — the computationally expensive phase that requires GPUs and weeks of time. *Inference* is using a trained model to produce predictions or generate text — much cheaper, and what happens when you type a prompt into ChatGPT.

Parameter. A single number inside a model that is adjusted during training. Neural networks are defined by their parameter values: change the parameters and you change what the model does. GPT-3 has 175 billion parameters. The goal of training is to find parameter values that make the model perform well.

Hyperparameter. A setting that controls the training process itself, as opposed to a parameter learned during training. Examples include the learning rate (how large each gradient descent step is), the number of layers in the network, and the batch size. Hyperparameters are set by the engineer before training begins and are not updated by gradient descent.

Core Machine Learning Concepts

Model. A parameterized mathematical function that maps inputs to outputs. A linear regression line is a model with two parameters. GPT-3 is a model with 175 billion parameters. In both cases, the model is defined by its parameter values and produces a prediction when given an input.

Training. The process of finding parameter values that minimize a loss function on a dataset. Training proceeds by gradient descent: compute the loss on a batch of examples, compute the gradient of the loss with respect to all parameters via backpropagation, and update each parameter by a small step in the direction that reduces the loss. Training a large language model requires weeks of computation on thousands of GPUs.

Epoch. One complete pass through the entire training dataset. Training typically runs for multiple epochs, allowing the model to see each training example many times. The number of epochs is a hyperparameter.

Batch. A subset of the training data used to compute one gradient descent update. Rather than computing the gradient on the entire dataset at once (expensive) or on one example at a time (noisy), training uses randomly sampled batches of typically 32 to 512 examples. This variant is called *stochastic gradient descent* (SGD).

Learning Rate. A hyperparameter that controls the step size in gradient descent. If the learning rate is too large, updates overshoot the minimum of the loss function. If too small, training converges very slowly. Choosing and scheduling the learning rate is one of the most important practical decisions in training a neural network.

Loss Function. A function that measures how wrong a model's predictions are on a given dataset. The goal of training is to minimize the loss function. Common choices include mean squared error (MSE) for regression and cross-entropy for classification. The loss function is the mathematical definition of what "better'' means for a particular task.

Overfitting. A condition where a model performs well on the training data but poorly on new, unseen data. An overfit model has memorized the training examples — including their noise and idiosyncrasies — rather than learning the underlying pattern. Detected by comparing performance on a held-out test set to performance on the training set.

Train/Test Split. The practice of dividing available data into a training set (used to fit the model) and a test set (used to evaluate how well the model generalizes to new data). The test set is never shown to the model during training. A typical split is 80% training and 20% test. This is the machine learning equivalent of using a sample to estimate a population parameter, then verifying the estimate on new data.

Gradient Descent. The optimization algorithm used to train neural networks. Starting from random parameter values, gradient descent repeatedly computes the gradient of the loss function with respect to all parameters and updates each

parameter by a small step in the direction that reduces the loss. The update rule is $\theta \leftarrow \theta - \alpha \cdot \nabla_\theta L$, where α is the learning rate and $\nabla_\theta L$ is the gradient.

Backpropagation. The algorithm that computes the gradient of the loss function with respect to every parameter in a neural network. Backpropagation applies the chain rule from calculus layer by layer, starting from the output and working backward to the inputs. It is the computational engine that makes gradient descent tractable for networks with millions or billions of parameters.

Neural Network Architecture

Neuron (Artificial). The basic computational unit of a neural network. A neuron takes a vector of inputs, computes a weighted sum, adds a bias term, and passes the result through an activation function to produce a single output value. A single neuron with a sigmoid activation is mathematically equivalent to logistic regression.

Layer. A collection of neurons that all receive the same inputs and produce outputs that are passed to the next layer. Neural networks are organized as a sequence of layers: an input layer, one or more hidden layers, and an output layer. The "depth'' of a network refers to how many layers it has.

Feedforward Network. A neural network in which information flows in one direction — from input to output — with no cycles or loops. Each layer's output becomes the next layer's input. Also called a multilayer perceptron (MLP). The feedforward network is the foundation of all more complex architectures.

Activation Function. A nonlinear function applied elementwise to the output of each neuron. Without activation functions, a neural network of any depth would collapse to a single matrix multiplication and lose all expressive power. Common choices include the sigmoid function (used in the output layer for binary classification), ReLU (used in hidden layers of most modern networks), and softmax (used in the output layer for multiclass classification).

Sigmoid. The activation function $\sigma(z) = 1/(1+e^{-z})$, which maps any real number to a value between 0 and 1. Useful as an output activation for binary classification because the output can be interpreted as a probability. The derivative of the sigmoid has an elegant form: $\sigma'(z) = \sigma(z)(1 - \sigma(z))$.

ReLU (Rectified Linear Unit). The activation function $\text{ReLU}(z) = \max(0, z)$, which outputs the input unchanged if positive and zero otherwise. The standard activation function for hidden layers in modern networks because it avoids the vanishing gradient problem that affects sigmoid in deep networks, and is computationally cheap to evaluate.

Softmax. A function that converts a vector of real numbers into a probability distribution: each output is non-negative and all outputs sum to 1. Used as the output activation for multiclass classification and in the attention mechanism. For a vector $\mathbf{z}$, $\text{softmax}(\mathbf{z})_i = e^{z_i} / \sum_j e^{z_j}$.

Cross-Entropy Loss. The standard loss function for classification problems. It measures the difference between the model's predicted probability distribution and the true class labels. Minimizing cross-entropy is equivalent to maximizing the likelihood of the training data under the model — a connection to maximum likelihood estimation in statistics. For binary classification with prediction $\hat{y}$ and true label y: $L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$.

Language Model Concepts

Token. The basic unit of text that a language model processes. A token is typically a word or a word fragment — common words like "the'' are single tokens; rarer words may be split into multiple tokens. GPT-3 uses a vocabulary of about 50,000 tokens. Text is converted to a sequence of integer token indices before being fed to the model.

Tokenizer. The component that converts raw text into a sequence of tokens. Tokenizers are trained separately from the language model, using algorithms that balance vocabulary size against the average number of tokens per word. The same text can produce different token sequences under different tokenizers.

Embedding. A dense vector representation of a token, learned during training. The embedding matrix maps each token index to a vector in $\mathbb{R}^d$ (where d is typically in the hundreds or thousands). Words with similar meanings end up with similar embedding vectors as a consequence of training, not by design. The embedding matrix is a parameter matrix updated by gradient descent like any other.

Context Window. The maximum number of tokens a model can process at once. GPT-3 has a context window of 2,048 tokens (roughly 1,500 words). Modern models have extended this to 100,000 tokens or more, allowing them to process book-length documents. Everything outside the context window is invisible to the model.

Next-Token Prediction. The training objective of most large language models. Given a sequence of tokens, the model is trained to predict the probability distribution over the next token. This simple objective, applied to enormous amounts of text, is sufficient to produce models capable of answering questions, writing code, translating languages, and reasoning about mathematics.

Prompt. The input text provided to a language model at inference time. The model generates a continuation of the prompt by repeatedly sampling the next token from its output distribution. Everything the model "knows'' about the current task is communicated through the prompt — there is no other input channel.

Hallucination. A confident but incorrect output from a language model. Because the model generates text by sampling from a learned probability distribution rather than retrieving facts from a database, it can produce plausible-sounding but false statements. Hallucination is a fundamental property of generative models; it can be reduced but not eliminated by training.

Temperature. A hyperparameter that controls the randomness of text generation.

At temperature 0, the model always selects the most probable next token (deterministic but repetitive). At higher temperatures, less probable tokens are selected more often (more creative but less coherent). Temperature scales the logits before the softmax: $\text{softmax}(\mathbf{z}/T)$, where T is the temperature.

Transformer Architecture

Attention. A mechanism that allows each token in a sequence to gather information from other tokens, weighted by learned relevance scores. Attention is the core innovation of the transformer architecture. Unlike a feedforward layer (which applies fixed weights to fixed positions), attention weights are computed dynamically from the content of the input.

Query, Key, Value (Q, K, V). The three learned projections used in the attention mechanism. The query represents what a token is looking for; the key represents what a token offers; the value represents what a token contributes to other tokens. Relevance between two tokens is measured by the dot product of their query and key vectors. The output for each token is a weighted sum of all value vectors, with weights determined by query-key similarity.

Attention Head. One instance of the attention mechanism operating with its own set of query, key, and value projection matrices. A transformer layer uses multiple attention heads in parallel (multi-head attention), each learning to attend to different kinds of relationships between tokens. GPT-3 uses 96 attention heads per layer.

Transformer Block. The repeating architectural unit of a transformer. Each block consists of a multi-head attention sublayer followed by a feedforward sublayer, with residual connections and layer normalization applied after each. GPT-3 stacks 96 transformer blocks.

Residual Connection. A shortcut that adds a layer's input directly to its output: $\text{output} = F(\mathbf{x}) + \mathbf{x}$. Residual connections allow gradients to flow directly through many layers during backpropagation, making it possible to train very deep networks without the vanishing gradient problem.

Positional Encoding. A vector added to each token's embedding to encode its position in the sequence. Because the attention mechanism computes all pairwise similarities simultaneously (with no inherent sense of order), positional information must be injected explicitly.

Layer Normalization. A technique that normalizes the activations within each layer to have zero mean and unit variance. Applied after each sublayer in a transformer block, it stabilizes training and reduces sensitivity to the choice of learning rate.

Training Methods and Fine-Tuning

Pre-training. The initial training phase in which a model is trained on a large general-purpose dataset (typically the text of the internet) using next-token prediction as the objective. Pre-training is computationally expensive and produces a model with broad language understanding. GPT-3's pre-training is estimated to have cost around $4 million in compute at 2020 prices.

Fine-tuning. Further training of a pre-trained model on a smaller, task-specific dataset. Fine-tuning adjusts the model's parameters to improve performance on a particular task (answering questions, writing code, following instructions) while retaining the general capabilities acquired during pre-training. Fine-tuning is much cheaper than pre-training because it starts from a good initial parameter configuration.

RLHF (Reinforcement Learning from Human Feedback). A training technique used to align language models with human preferences. After pre-training, human raters compare pairs of model outputs and indicate which is better. A separate reward model is trained on these preferences, and the language model is then fine-tuned using reinforcement learning to maximize the reward model's score. RLHF is the primary technique responsible for the difference between a raw language model (which predicts text statistically) and a helpful assistant (which follows instructions and avoids harmful outputs). ChatGPT and Claude were both developed using variants of RLHF.

Mixture of Experts (MoE). An architectural technique in which a model contains many parallel "expert'' subnetworks, with a learned gating function that routes each input to only a small subset of experts. MoE allows a model to have far more total parameters than a dense model of equivalent compute cost, because most parameters are inactive for any given input. DeepSeek-R1 has 671 billion total parameters but activates only about 37 billion per token.

Scaling Laws. Empirical relationships between model performance and the three main resources used in training: model size (number of parameters), dataset size (number of training tokens), and compute (number of floating-point operations). Performance improves predictably as a power law in each of these quantities. Scaling laws allow researchers to predict how much a model will improve before training it, and to determine the optimal allocation of a fixed compute budget between model size and data.

Note: These handouts were developed collaboratively with Claude Sonnet 4.6 (Anthropic) through an iterative conversation. Source files are written in Markdown with LaTeX equations and converted to PDF using Pandoc and XeLaTeX.